

A. **Chinese Remainder Theorem.** Now we solve systems of congruences like:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

An exhaustive guess-and-check yields one solution: $x =$

Are there others? We perform a general analysis of these sorts of systems.

Chinese Remainder Theorem. Let $n_1, n_2, \dots, n_r$ be positive integers that are

Let $a_1, a_2, \dots, a_r$ be integers. Then the system

$$x \equiv a_1 \pmod{n_1}$$

$$x \equiv a_2 \pmod{n_2}$$

$$\vdots$$

$$x \equiv a_r \pmod{n_r}$$

has a solution, and it is unique modulo:

Proof. First we show existence.

The key is to work with products of all moduli except one.

$$N_k \stackrel{\text{def}}{=} \frac{n}{n_k} =$$

First observe that, since n_k has no common prime factors with the other moduli

So the following congruence has a solution, which we call x_k .

We will show that the following solves the original system of congruences:

$$x =$$

Modulo n_k we verify

$$x \equiv$$

We next prove that the solution is unique modulo the product of moduli.

Let x and x' be two solutions to the system. Then for each k in $[1 \dots r]$ we have

$$x - x' \equiv$$

Because the n_k are pairwise relatively prime, $n_1 n_2 \cdots n_r \mid$

Hence x and x' are congruent modulo $n = n_1 n_2 \cdots n_r$. $\square$

Example 1. Find all numbers that leave remainders of 2, 3, 2 when divided respectively by 3, 5, 7.

This is a problem that was posed by Sun Zi of China in the third century CE.

Solution. We search for such numbers x by setting up a system.

Observe that:

Next we construct

$$n =$$

$$N_1 =$$

$$N_2 =$$

$$N_3 =$$

Next we solve the congruences $N_k x_k \equiv a_k \pmod{n_k}$.

The unique solution modulo n is the sum of the $N_k x_k$.

$$x \equiv$$

Thus each solution x is uniquely of the form

$$x =$$

B. **Decomposition of Congruences.** We can decompose singular congruences.

Theorem. Let $n_1, n_2, \dots, n_r$ be positive integers that are

and let $n =$

Let a and b be integers. Then the congruence

$$a \equiv b \pmod{n}$$

is equivalent to the system of congruences

Proof. Observe that $a \equiv b \pmod{n}$ if and only if

which, since the n_k are pairwise relatively prime, is equivalent to

which is equivalent to the given system of congruences.

□

Example 2. Use that $276 = 3 \cdot 4 \cdot 23$ and the Chinese remainder theorem to solve

$$14x \equiv 10 \pmod{276}$$

Solution. Since

we can decompose into the equivalent system

Observe that:

Next we construct

$$n =$$

$$N_1 =$$

$$N_2 =$$

$$N_3 =$$

Next we solve the congruences $N_k x_k \equiv a_k \pmod{n_k}$.

The unique solution modulo n is the sum of the $N_k x_k$.

$$x \equiv$$

Thus each solution x is uniquely of the form

$$x =$$